

Tanuj Dontaraju

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EDUCATION

University of Delaware

B.S. in Artificial Intelligence Engineering

GPA: 3.8/4.0

Relevant Coursework: Calculus I–III, Data Structures, Discrete Mathematics I, Data Science I, Engineering Mathematics I, Introduction to Systems Programming, Introduction to Digital Systems

EXPERIENCE

University of Delaware - DIR Lab

Aug. 2026 – Present

Incoming Research Assistant

Mercor

Aug 2025 – Jan 2026

AI / Software Engineering Contributor

- Curated and labeled datasets across image and text domains to support machine learning model training.
- Preprocessed and validated data to improve dataset quality and consistency for training pipelines.
- Worked within structured workflows to maintain high data quality standards for AI development.

PROJECTS

StudyLens (Personalized RAG Chatbot)

Python, Streamlit, LangChain, ChromaDB, LLM APIs · <https://github.com/TanujDontaraju/StudyLens>

- Developed a Retrieval-Augmented Generation (RAG) chatbot using Python and LangChain for personalized course materials.
- Built a PDF/text ingestion pipeline with ChromaDB for local vector search.
- Created a Streamlit interface supporting OpenAI, Gemini, and Anthropic LLMs.

AI Race Strategy Simulator (In Progress)

Python, Data Modeling, Simulation Systems · <https://github.com/TanujDontaraju/F1-Race-Strategy>

- Developing a data-driven simulation system to model Formula 1 race outcomes using historical lap times, tire degradation, and track-specific variables.
- Building a lap-by-lap simulation engine incorporating probabilistic events such as pit strategies, safety cars, and performance variability.
- Designing a strategy optimization module to evaluate and compare multiple race strategies (e.g., undercut vs overcut).

SafeCircle (Mobile Safety Application)

Team Project (3 contributors)

TypeScript, REST APIs, Geolocation, AI Integration · <https://github.com/TanujDontaraju/SafeCircle>

- Developed a mobile safety application with real-time location tracking and emergency communication features.
- Integrated speech-to-text and generative AI APIs to process and summarize audio data during emergency scenarios.
- Built backend endpoints for location tracking, audio processing, and alert systems with secure API handling.

Steganography Project (Pixel Parse)

Python, Data Encoding, Image Processing · <https://github.com/TanujDontaraju/Steganography-Project>

- Designed a full-stack steganography system for hiding and retrieving secret messages inside images using LSB encoding.
- Implemented ASCII-to-binary conversion pipelines and pixel-level data manipulation techniques.
- Built a state-driven web interface for encryption and decryption workflows.

SKILLS

Programming Languages: Python, JavaScript, Java, HTML, CSS

Technologies: LangChain, Streamlit, Google Gemini API, PyTorch, Retrieval-Augmented Generation (RAG), TensorFlow, Git